

Molecular virulence mechanism of phospholipase C from *Pseudomonas aeruginosa*

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SUMMARY

Bacterial phospholipases form a structurally heterogeneous group of lipolytic enzymes that hydrolyze one or more ester bonds in their substrates. Many of these enzymes are essential virulence factors required for the efficient infection of human or animal host cells. *Pseudomonas aeruginosa* secretes two virulent glycerophospholipases of type C (PLC N and PLC H), the latter of which has hemolytic activity on red blood cells. Cryo-electron microscopy and X-ray crystallography analysis showed that both proteins have two domains with similar tertiary structures, conserved active site residues, and surprisingly the phosphorylation of the active site threonine that acts as the nucleophile in lipase activity. Docking analysis suggests that the distinct substrate preference of PLC N for phosphatidylserine and PLC H for phosphatidylcholine is mainly caused by electrostatic differences in the substrate binding pocket. Further time-resolved studies using atomic force microscopy (AFM) showed how PLC H can damage host membranes by forming large pores in lipid bilayers within minutes.

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Dear Editor,

Pseudomonas aeruginosa, is a major public health concern due to its multi drug resistance (MDR) and pathogenicity¹. These Gram-negative bacteria possess a vast range of known virulence factors, including phospholipases^{2,3}. *P. aeruginosa* secretes four different PLCs of which the hemolytic PLC H is the most abundant and major virulence factor⁴. It hydrolyzes phosphatidylcholine (PC), sphingomyelin (SM), and phosphatidylserine (PS), contributes to nutrient acquisition and osmoprotection, and exhibits ceramide-dependent SM synthase activity^{5,6,7}. By degrading lung surfactant, PLC H promotes bacterial survival, biofilm formation³, immune responses and host cell disruption^{8,9}. It forms a cytosolic complex together with its cognate chaperone PLC R, which has two different mature forms, PLC R2 (19 kDa), and PLC R1 differing by 36 amino acids at the C-terminus. PLC R1 was suggested to play a role in outer membrane secretion, while PLC R2 might assist in inner membrane transport. The non-hemolytic PLC N and PLC B cleave PS and PC, and phosphatidylethanolamine (PE) respectively^{2,3}. Unlike lipases in Gram-positive bacteria such as *Clostridium perfringens*, which belong to the superfamily of acid phosphatases and show no structural or sequence homology to PLCs from *Pseudomonas*, they do not require Zn²⁺ ions to function. Based on a previous study on 414 clinical isolates^{11,12} we identified PLC H and PLC N as relevant pathogenic factors and investigated in the present study the structure and function of the purified hemolytic PLC H and the non-hemolytic PLC N using different activity assays, biophysical methods, and high-resolution structural analysis on wild-type and some mutant proteins to understand the molecular mechanism of PLC virulence of *P. aeruginosa*.

The genes of PLC N, PLC H, and its cognate chaperone PLC R2 (19 kDa) were heterologously expressed in *E. coli*, purified (**Fig. S1 and S2**) and their identity confirmed by mass spectrometry (**Fig. S3 and S4**). Mass photometry and dynamic light scattering (DLS) showed monomeric PLC N in solution and 1:1 stoichiometry of PLC H/R2 complex (**Fig. S5 and S6**); the latter confirmed by size-exclusion chromatography coupled with small-angle X-ray scattering (SEC-SAXS) and on-line multi angle light scattering (MALS) (**Fig. 1A, Fig. S7 and Table S1**). In agreement with *ab-initio* determination of the shape topology (**Fig. S8**) and by rigid body modeling of the AlphaFold structures to the SAXS data, the chaperone was located on the N-terminal domain of PLC H, close to the active site (**Fig. 1B**).

In contrast to PLC N, the complex PLC H/R2 displayed β -hemolysis of red blood cells in sheep blood agar plates (**Fig. 1C**). Both phospholipases were cytotoxic to HeLa cells in a 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT) assay resulting in a lethal dose (LD50) of 0.005 μ g/ μ L for PLC H/R2 complex and 5.6 μ g/ μ L for PLC N, respectively (**Fig. 1D-F**). To compare enzymatic activities, we conducted a lipase assay with p-nitrophenyl phosphorylcholine as the substrate (**Fig. S9**). The specific enzymatic activity of PLC H/R2 and PLC N was determined to be 3.8×10^5 and 3.3×10^3 nmol/min.mg, respectively, and thus differs by a factor of 10^2 . The cytotoxicity of PLC H/R2 however, was about 10^5 times higher than that of PLC N suggesting that cell death induced by these enzymes occurs via different mechanisms.

Since the interaction of PLC H/R2 with host membranes is poorly understood we monitored the structural integrity and dynamics of targeted lipid bilayers using atomic force microscopy (AFM) (**Fig. 1G-H, Fig. S10-S14**). The formation of holes in DMPC bilayers was observed after 7 min. Their diameter increased over time, from ca. 0.1 to 2.5 μ m with a typical depth of 3.5 to 4.0 nm, affecting 80 % of the monitored area after 30 min. At 15 to 25 minutes incubation, we observed rim formation at the pores (**Fig. 1G and Fig. S15**) with a typical width of 15 to 25 nm. The formation of the rim structure (**Fig. S16**) added 3 nm to the observed depth of the holes. Within 30 min the pores flattened and only a 3 nm extension of the rim structure above the membrane surface remained observable (**Fig. 1H**).

Cryo-electron microscopy (cryo-EM) analysis showed three distinct molecular species with uniform shape (**Fig. 2A**, and **Fig. S17**) which we assigned as PLC H monomer, dimer, and trimer. Each oligomer had a distinct structure different from the shape observed in SAXS experiments, indicating that the enzyme lost its cognate chaperone during cryo-EM sample preparation. Since the particles adopted a preferred orientation on the EM grid surface, we collected data at three different tilt angles (0, 30, and 45 degrees) as summarized in **Table S2**. The 3D reconstruction of the monomer and trimer particles resulted in three distinct maps representing the PLC H monomer with dimensions of approximately 60 x 48 x 66 Å (**Fig. S18**), and two trimers of distinct conformations (**Fig. S19**), both having a horseshoe-like shape, but one slightly more open than the other. In the oligomers, the molecules were bound head to tail showing the capability to facilitate specific protein-protein binding (**Fig. 2A**) as required for the pore formation during membrane lysis as mentioned above.

Since the oligomer subunits adopted an almost identical structure, we applied map averaging to improve the model of PLC H (see Methods). The starting structure used for positioning and further refinement was derived from the AlphaFold model. **Table S3** summarizes the refinement statistics of PLC at 3.54 Å resolution with an FSC of 0.143, including 692 of the protein's 730 residues. The 14 N-terminal residues (6His-tag and a TEV cleavage site), and 38 N-terminal residues of PLC H were too flexible and could not be resolved.

The PLC H structure consisted of an N-terminal and a C-terminal domain (**Fig. 2B**). Seven conserved residues - E60, N61, H120, H321, D357, E358, and the nucleophile T178, - were located in a surface cavity of the N-terminal domain (residue 1 – 470) with mixed alpha and beta structure identifiable as part of the acid phosphatase superfamily^{13,14}. The lipase active site contained the predicted nucleophile T178 in its phosphorylated state as indicated by the cryo-EM map (**Fig. 2C**). Phosphorylation was consistently observed across different protein preparations and confirmed by LC-MS/MS (**Fig S3**). It should be noted that the nucleophile at the active side of both PLCs appear to be fully phosphorylated, suggesting that proteins isolated from bacteria are in a stable state. Both *wild type* proteins displayed enzymatic activity (**Fig. S9**), and substitution of the nucleophile T178 with alanine (T178A, T176A in PLC N) almost completely abolished lipase activity in both enzymes (T178A: 445, and T176A 144 nmol/min.mg)¹⁵, suggesting that the phosphorylation of the active side threonine is part of the reaction mechanism. Since experiments were conducted with purified protein future studies should further investigate the physiological significance of phosphorylation.

The active site of PLC H displayed a negative surface potential (**Fig. S20**) suitable for stabilization of positively charged lipid headgroups e.g. choline. The active site contained a Ca²⁺-ion bound amongst others to the phosphorylated nucleophile T178 (**Fig. 2C**). Neither EDTA, nor the presence of Ca²⁺-, or Mg²⁺-ions (each 1 mM) abolished the enzyme activity (EDTA, Ca²⁺, Mg²⁺ ~ 3.3 × 10³ nmol/min.mg for PLC N and 3.8 × 10⁵ for PLC H), whereas additional Zn²⁺-ions completely suppressed enzymatic cleavage activity (**Fig. S21**). The latter can be explained by assuming that Zn²⁺-ions in PLC N are coordinated by H296 and H118 (in PLC H these are H321 and H120, resp.) which affects the Ca²⁺ binding and also influences the geometry of the active side residues S178 (S176) and D333 (D357).

The closest known structure similar to this domain was the acid phosphatase AcpA from the zoonotic pathogen *Francisella tularensis* with a sequence similarity of 22.1 %, according to DALI¹⁵, an RMSD of 1.96 and 1.81 Å for PLC H and PLC N resp., and seven conserved active site residues (**Fig. S22**). The C-terminal PLC H domain folds into a beta-sandwich (4+3 β stranded sheets) structure with Ig/C2 fold (S-type) without a Ca²⁺ binding site. This typical recognition domain fold is known to bind highly diverse classes of molecules including proteins, carbohydrates, and lipids e.g. in phospholipase C of *C. perfringens* (α-toxin).

With the aim of understanding how different substrate specificity is achieved^{8,10,16} we analyzed the PLC N structure by X-ray crystallography at 1.6 Å resolution (**Fig. S23**) using the AlphaFold model for molecular replacement and refined to a final R_{free} of 0.152 (**Table S4**). The asymmetric unit contained a crystallographic PLC N dimer with proper 2-fold symmetry (**Fig.**

S24). The N-terminal and C-terminal domain of PLC N adopt the same fold as PLC H (**Fig. S25**), corroborated by the similarity of secondary and tertiary structure (**Table S5**). Further nano-DSF and CD experiments showed that the PLC H/R2 complex is thermally more stable than PLC N (**Fig. S26 and S27**)

Within the active site of PLC N phosphorylated T176 and two ions were identified. After testing the lipase activity in the presence of different cations we identified and verified the presence of Ca^{2+} -ions through temperature factor and occupancy refinement, taking into account geometric and distance criteria. Presence of a second Ca^{2+} -ions in PLC N can be attributed to binding by the non-conserved negatively charged residue D241. Overall, the two proteins exhibited a significant difference in the charge distribution and the shape of the active site which likely contributed to the binding of the second Ca^{2+} -ions (**Fig. S20**). The Ca-binding site consisted of four near ligands (**Fig. S28**) and four additional sidechains in van-der-Waals distance that contributed to binding.

Compared to PLC H the C-terminal domain of PLC N prominently featured a positively charged patch (R and L rich) whereas in the N-terminal domain negative charges were concentrated in the active site (**Fig. S20**). This negative charge distribution could help target the positively charged choline group towards the active site and may contribute to the differing substrate affinities and enhance activity. The accumulation of positive charges on the C-terminal domain disfavored interaction with hydrophobic lipids apart from negatively charged headgroups, and pointed at membrane surface interaction, whereas the comparatively uncharged C-terminal domain of PLC H allowed for membrane interaction as observed in the AFM experiments.

Taken together, the two *P. aeruginosa* PLCs investigated here share a common fold, suggesting common descent from the acid phosphatase superfamily. Both enzymes have conserved active site residues, yet differ markedly in activity. PLC H displays strong hemolytic properties verified by AFM indicating protein stabilized pores in membranes which in turn might provide efficient access to host resources.

We analyzed the possible interactions of PLC H and PLC N with their respective substrates by molecular docking with and without Ca^{2+} -ions and nucleophile phosphorylation using the program Autodock Vina 1.2. The most likely near attack conformations (NAC) of the respective substrates were identified by their distances to the nucleophile (**Table S6**). These results suggest that lipase substrates PC and PS and synthetase substrate ceramide (CM) form stable complexes with PLC H. The lower binding energies support the higher activity and virulence observed for PLC H compared to PLC N. The different specificities are likely due to individual shapes of the active sites. Indeed, PLC H showed an adequate substrate binding pocket with a less crowded and prominent cavity, while the active site of PLC N is rather shallow (**Fig. S20**) and possibly less restraining on the conformation of the fatty acids in the binding substrate.

The docking model of PC to PLC H suggested the quaternary amino group located at the center of the conserved negatively charged side chains (E60, E121, D357, E358, D360) in the active site, suggesting that the phosphocholine group was positioned mainly by electrostatic forces and hydrogen bonds. The substrate conformation (**Fig. 2D**) was stabilized by interactions with E60, H120, E121, T178, N203, Y264, D357, E358, and D360, and the main chain N of residue N203. Together, these residues were holding the PC in a suitable NAC. This conformation was further stabilized by hydrogen bonds with active site residues E60, T178, and N203 resulting in low binding energies (**Table S6**). In the docking complex of PLC N with PC (**Fig. 2G**) the substrate phosphate group was bound by a hydrogen bond to D204 implying the presence of a proton. In the docking structure, the distance between the nucleophile T176 and substrate phosphate measured about 5 Å (**Fig. 2G**). Here, the low binding energies also suggested the presence of a stable complex (**Table S6**).

The docking models with PS suggested the serine moiety oriented into the active site cavity of PLC N forming six hydrogen bonds in total (**Fig. 2H**). The phosphate of the lipid was bound by N201 and D204. The nitrogen of the serine moiety formed hydrogen bonds with E57, T176, D241 of PLC N. For the PLC H PS complex the serine moiety was oriented by hydrogen bonds

with N179 and D265. The phosphate of the PS was stabilized by hydrogen bonds with T178 and N179 of PLC H. Other active site residues in van-der-Waals distance (<5 Å) assisted the NAC (**Fig. 2E**).

The docking models for ceramide with the phosphorylated protein suggested that in PLC H a NAC is achieved by hydrogen bonds formed with Y249, Y316, and H321 (**Fig. 2F**). For PLC N, the orientation of ceramide was achieved by hydrogen bonds formed with N201, D204, and the phosphate at T176 (**Fig. 2I**). The PLC N structure without nucleophile phosphorylation did not yield any NAC model, stressing the binding contribution of the phosphate and the lack of other strong interactions of PLC N's active site with ceramide. Based on the required active site cavity volumes binding of ceramide and PC could not be adopted simultaneously.

Overall, the substrate docking analysis supported the observed substrate specificities without PLC R2 playing a role in this process. For spatial reasons two consecutive reaction steps are therefore required in the synthetase reaction which would require the formation of a covalent bond between phosphocholine and the nucleophile T178 in PLC H.

Red blood cells, rich in PC and SM, are especially susceptible to PLC H, whereas HeLa cells used in the MTT-assay contain less favorable substrates for PLC N. The stronger activity of PLC H against RBCs thus explains its pronounced hemolytic phenotype. Structural docking suggested that variations in non-conserved residues alter shape and electrostatics of the active site, thus influencing substrate selectivity. PLC H additionally exhibits sphingomyelin synthase activity, linking ceramide and phosphocholine, though this is not essential for virulence. The presence of the phosphorylated lipase nucleophile threonine suggests that PLC H performs lipase activity and retains the phosphocholine bound to the nucleophile which could be transferred in a second step after binding the substrate ceramide. Luberto and colleagues showed the reversibility of this reaction^{13,14} indicating that a net production of sphingomyelin requires ceramide and phosphocholine, which can be obtained by PLC H from the phosphatidylcholine cleavage. We speculate that the synthetase activity is always preceded by a lipase activity step. In the absence of ceramide, the lipase activity would be inhibited or diminished by the covalent modification of the nucleophile. Therefore, the intermediate would have to hydrolyze to regain lipase activity (**Fig. S29**). The mass spectrometric and structural characterization of the protein required that hydrolysis must proceed in two steps, the loss of choline followed by slower dephosphorylation. Indeed, the preferential cleavage of the diester bond of the covalent intermediate between the nucleophile threonine and phosphate would be detrimental to synthetase activity (**Fig. S29**).

We conclude that neither the synthetase activity nor the reaction products of the lipase activity were required for the increased virulence of PLC H compared to PLC N. Our results are consistent with *in vivo* mice studies showing restored virulence upon PLC H complementation of the *P. aeruginosa* knockout with PLC H^{17,18}. Thus, we attribute the hemolytic activity to the ability of PLC H to form membrane pores in RBCs caused by PC digestion. Since we found PLC H to be insoluble in the absence of PLC R2, at least part of its functionality lies in the solubilization of PLC H before it binds to membranes and form pores. In the EM experiments, the sample preparation led to the dissociation of the chaperone and the formation of di- and trimeric PLC H complexes on the sample grid. We suspect that the oligomers are responsible for the observed membrane pores, although validation will require further investigation of these pore complexes. Our results suggest that the hemolytic activity is due to the oligomerization tendency of free PLC H, and consequently, the formation of membrane pores. The observed inhibition of lipase activity by Zn^{2+} aq highlights the potential of metal-binding sites, particularly Ca^{2+} and Zn^{2+} coordination, as promising targets for inhibitor development. It should be noted that Zn^{2+} can bind to lipids¹⁹ and hydrolyze phosphodiester bonds²⁰. Since Zn^{2+} -ions are not present in the PLC structures, they most likely inhibit enzyme activity by displacing active site Ca^{2+} -ions and interacting with the phosphodiester.

FOOTNOTES

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Conceptualization: J.L., M.K., and N.S. conceived this study, N.S., G.Y., M.L., U.S., J.C., S.C., and J.L. conducted formal analysis, N.S., U.S., A.T., J.C., S.S., G.Y., A.G., and M.L. performed experiments, J.L. and M.K. provided resources, G.Y., M.L., and J.L. performed data curation, N.S., U.S., and J.C. visualized the data, J. L. and M. K. supervised the project, N.S., J.L. and M. K. wrote, reviewed and edited the manuscript. All authors read and commented on the manuscript.

The authors declare they have no competing interests.

Structural data were deposited at the protein data bank with accession codes 9GYE for PLC H and 9GY2 for PLC N. Small Angle Scattering are available at the Small Angle Scattering Biological Data Bank (SASBDB) with accession code SASDV94.

Figure legends:

Figure1 Characterization, activity and DMPC bilayer membrane perforation by PLC H/R2

(A) Final merged SAXS profile and (B) AlphaFold model of PLC H/R2 showed occlusion of the enzyme active site by the its chaperone; NTD: N-terminal domain, CTD: C-terminal domain (PLC H in cyan, PLC R2 in yellow) For curve fitting parameters. supplementary Table S1. (C) PLC H/R2 (top) and PLC N (bottom) were incubated on 5 % sheep blood agar plates (70 µg/mL). PLC H/R2 showed β -hemolysis at 37°C after 16 h while PLC N had no lytic effect on red blood cells. The representative plate with blue arrow marks diffusion perimeter (biological replicates). (D) Outline of the MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl-2H-tetrazolium bromide) assay (E) Conversion of yellow MTT reagent into purple Formazan crystals in living cells (right: effect of staining on cells) (F) Cell death dose–response curves for PLC N (left) and PLC H/R2 (right). For curve fitting parameters. supplementary data. (G) Time course of membrane perforation as observed by AFM. White arrow marks locus of the same membrane perforation through time. (Scale bar 0.5 µm) (H) Time resolve mapping of membrane perforation after 7 min (top), 25 min (middle) and 28 min (bottom) by z-axis cross section (left) at selected perforation events (right, scan direction indicated by hyphenated line). AFM experiments were conducted as biological triplicates. (Scale bar 0.25 µm)

Figure 2. Structural analysis of PLC H and near attack conformers of substrates for PLC H and PLC N

(A) 2D-classification of species observed in single particle analysis showing head-to-tail trimers, dimers and monomers (left) and 3D-reconstruction of electron density for trimer (middle), and monomer at 3.54 Å (right) (B) N terminal (yellow) and C terminal domain (red) of PLC H with bound Ca-ion (green) (C) Electron density and LIGPLOT of the active site. Best docking results using Autodock Vina selected by attack distance to T178 (T176) are shown in

(D) to (I). PLC H is depicted in blue and PLC N in orange. (D) Unphosphorylated (T178) PLC H with preferred lipase activity substrate phosphatidylcholine (PC) (E) Unphosphorylated (T178) PLC H with lipase activity substrate phosphatidylserine (PS) (F) Phosphorylated (T178) PLC H with sphingomyelin synthetase activity substrate ceramide (CM) (G) Unphosphorylated (T176) PLC N with lipase activity substrate PC (H) Unphosphorylated (T176) PLC N with preferred lipase activity substrate PS (I) Phosphorylated (T176) PLC N with non-substrate CM.

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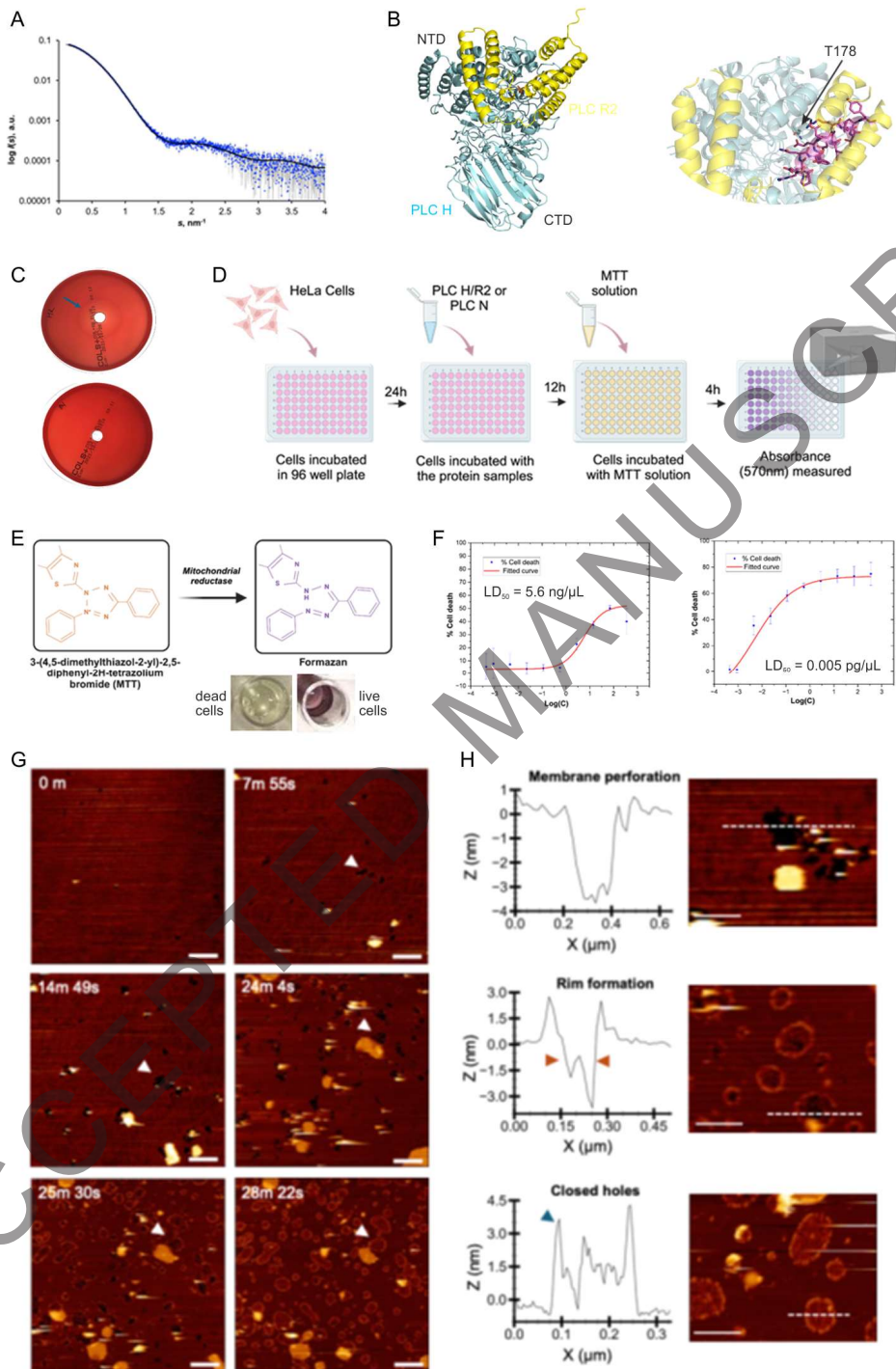


Figure1 Characterization, activity and DMPC bilayer membrane perforation by PLC H/R2

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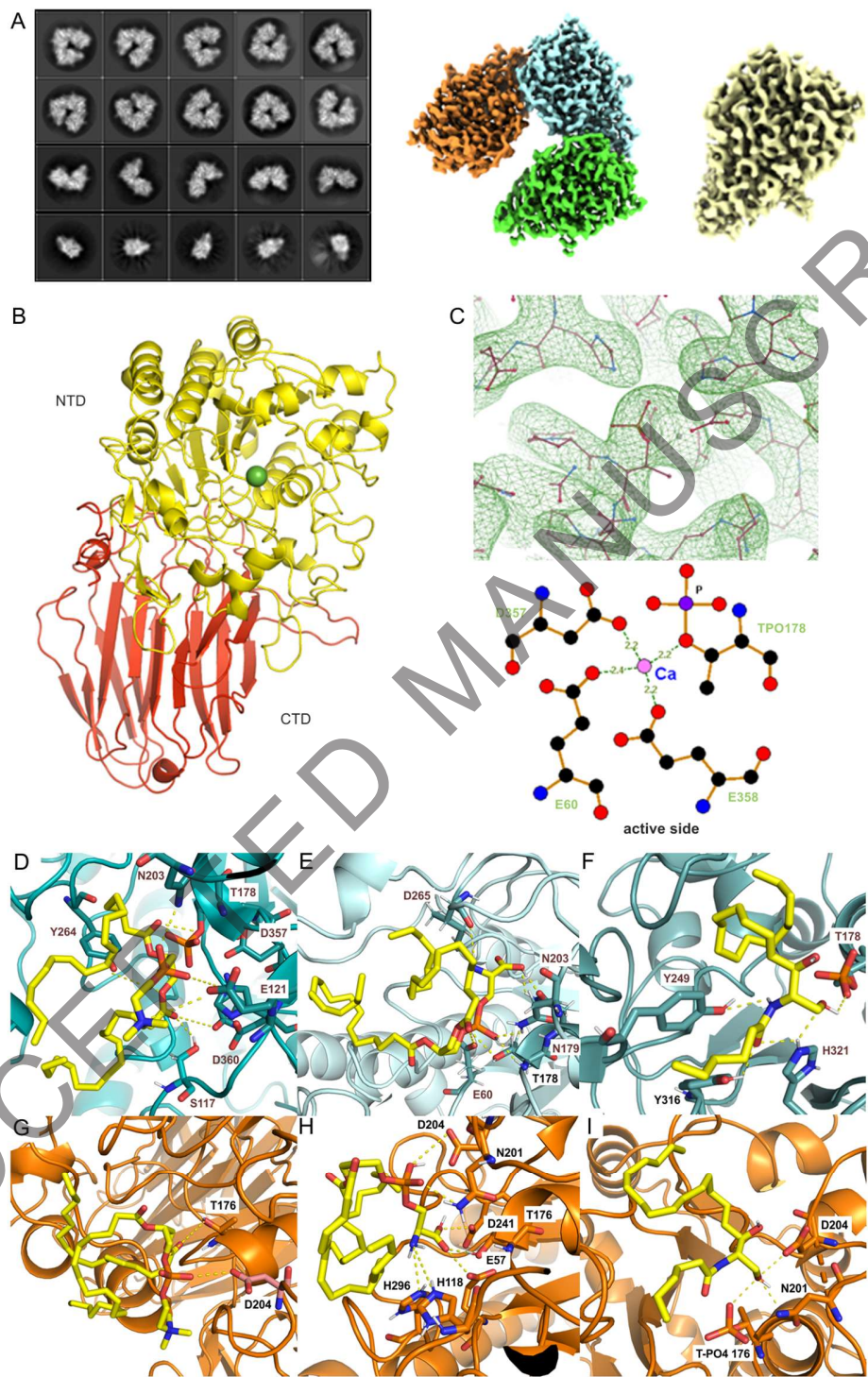


Figure 2. Structural analysis of PLC H and near attack conformers of substrates for PLC H and PLC N

(A) 2D-classification of species observed in single particle analysis showing head-to-tail trimers, dimers and monomers (left) and 3D-reconstruction of electron density for trimer (middle), and monomer at 3.54 Å (right) (B) N terminal (yellow) and C terminal domain (red) of PLC H with bound Ca-ion (green) (C) Electron density and LIGPLOT of the active site. Best docking results using Autodock Vina selected by attack distance to T178 (T176) are shown in (D) to (I). PLC H is depicted in blue and PLC N in orange. (D) Unphosphorylated (T178) PLC H with preferred lipase activity substrate phosphatidylcholine (PC) (E) Unphosphorylated (T178) PLC H with lipase activity substrate phosphatidylserine (PS) (F) Phosphorylated (T178) PLC H with sphingomyelin synthetase activity substrate ceramide (CM) (G) Unphosphorylated (T176) PLC N with lipase activity substrate PC (H) Unphosphorylated (T176) PLC N with preferred lipase activity substrate PS (I) Phosphorylated (T176) PLC N with non-substrate CM.